

Adarsh Yadav

AI Engineer | GenAI · LLM · RAG Systems

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SUMMARY

AI Engineer building production-grade **Large Language Model (LLM)** systems — **Retrieval-Augmented Generation (RAG)**, multi-agent workflows, and scalable GenAI backends — with a B.Tech from IIT Roorkee. Skilled in **prompt engineering**, **LLM evaluation**, **guardrails**, and **vector search**, and experienced in deploying event-driven serverless APIs on Azure at **99.9% uptime**. Strong CS fundamentals backed by 500+ DSA problems solved.

EDUCATION

Indian Institute of Technology Roorkee

2022 – 2026

B.Tech

CGPA: 7.99 / 10

EXPERIENCE

FnMathLogic Consulting Services

June 2026 – Present

GenAI-powered Education Platform

- **Engineered LLM-powered content generation pipelines** for automated, grade- and subject-specific daily lesson planning, incorporating contextual data, prompt engineering, and structured output generation.
- **Scaled high-volume LLM workloads** through Python multiprocessing, multithreading, and concurrent API execution, improving generation throughput and minimizing sequential inference bottlenecks.
- **Architected and deployed event-driven serverless APIs** with **Azure Functions** and **Azure Storage**, achieving **99.9% production uptime** while supporting scalable lesson-generation workloads.

PROJECTS

DocQuery — Multi-Tenant RAG Document Q&A Platform

Python, FastAPI, PostgreSQL, pgvector, Redis

- **Built** a production-style **multi-tenant RAG API** using FastAPI and async SQLAlchemy, enabling isolated document ingestion and semantic search with **LLM-generated, source-grounded answers**.
- **Engineered tenant isolation and API security** using **PostgreSQL Row-Level Security (RLS)**, API-key authentication with **SHA-256 key hashing**, and Redis-based **fixed-window rate limiting** for secure SaaS-style access control.
- **Designed an asynchronous document ingestion pipeline** with background processing, job-status tracking, **content-hash deduplication**, and Sentence Transformers embeddings for efficient **k-NN vector similarity search**.
- **Optimized retrieval performance** with a Redis **cache-aside architecture**, TTL-based expiration, and tenant-scoped cache invalidation; implemented **structured logging with request-ID correlation** using Structlog for end-to-end observability.

Multi-Agent Research Pipeline

Python, LangGraph, FastAPI, Groq, Tavily

- **Built** a multi-agent LLM research system using **LangGraph** with a Planner → Researcher → Writer ⇌ Critic workflow and a bounded revision loop to generate **sourced, verified reports** from user queries.
- **Developed** an evaluation harness to measure agent performance, demonstrating **80% source-grounding accuracy** and verifying that the Critic loop resolved flagged issues such as unsupported claims and information gaps; added **per-agent token usage and latency profiling**.
- **Hardened** the system with layered **input validation**, **LLM-based moderation**, and **provider/search fallbacks**, enabling graceful degradation during invalid inputs, moderation failures, or service outages.
- **Deployed** the frontend and FastAPI backend as separate services on **Render**, enabling independent deployment and scaling of application components.

SKILLS

Languages: Python, C++, SQL, JavaScript

AI / LLM Engineering: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Systems, LangGraph, LLM Evaluation & Benchmarks, LLM Guardrails, Embeddings, Sentence Transformers, Vector Search

Backend & Systems: REST APIs, FastAPI, Node.js, Express.js, System Design, Scalability, Microservices, Serverless (Azure Functions), Authentication, Async Programming

Databases & Caching: PostgreSQL, MongoDB, Redis, pgvector

Cloud & DevOps: Microsoft Azure AI, Docker, CI/CD, Git, GitHub, Linux, Observability (Structlog)

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS

- Secured **AIR 9162** in JEE Advanced 2022 among 1.6 lakh+ qualified candidates.
- Solved **500+ algorithmic problems** across Codeforces, CodeChef, and LeetCode, reflecting strong problem-solving and DSA fundamentals.